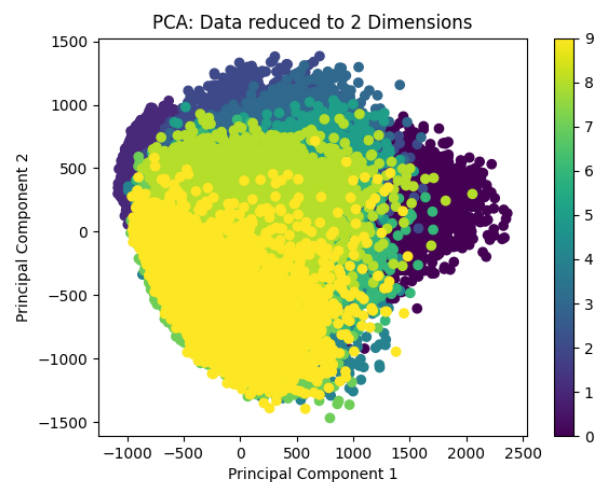
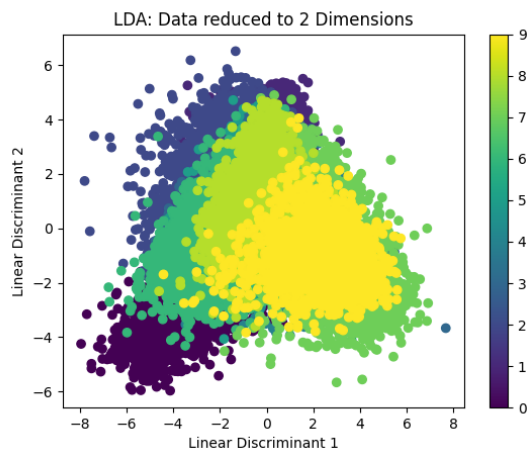


Machine Learning Lecture 5 Summary – Group 720

Perform classification for the entire MNIST dataset (the notebook uses the `mnist_all.mat` file attached to exercise 4) based on the algorithms introduced: Use LDA for dimensionality reduction to 2 or 9 dimensions, classify the dimension-reduced data and compare this classification performance with that of using PCA.

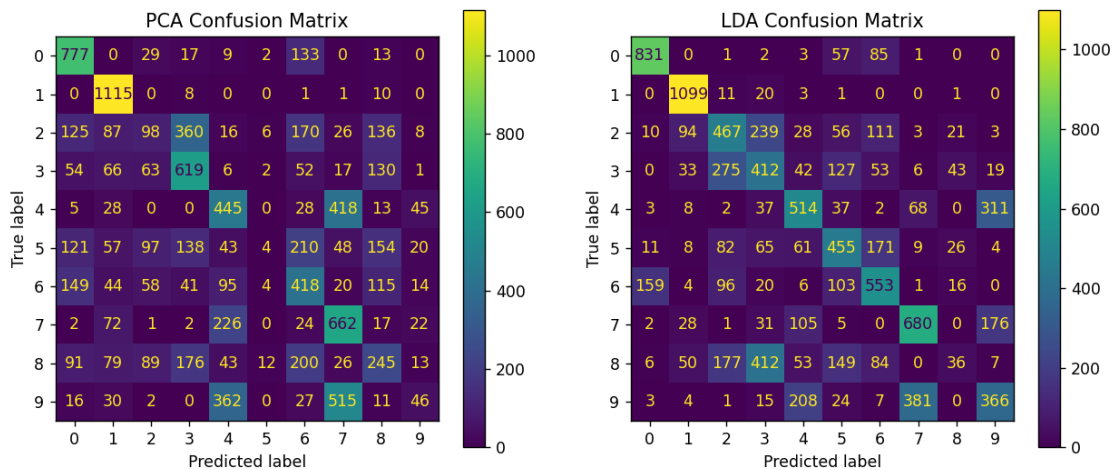
We first visualize the dimensionality reduction to 2 dimensions with both LDA and PCA:



Where we get the following accuracy and confusion matrices:

PCA Accuracy: 0.4429

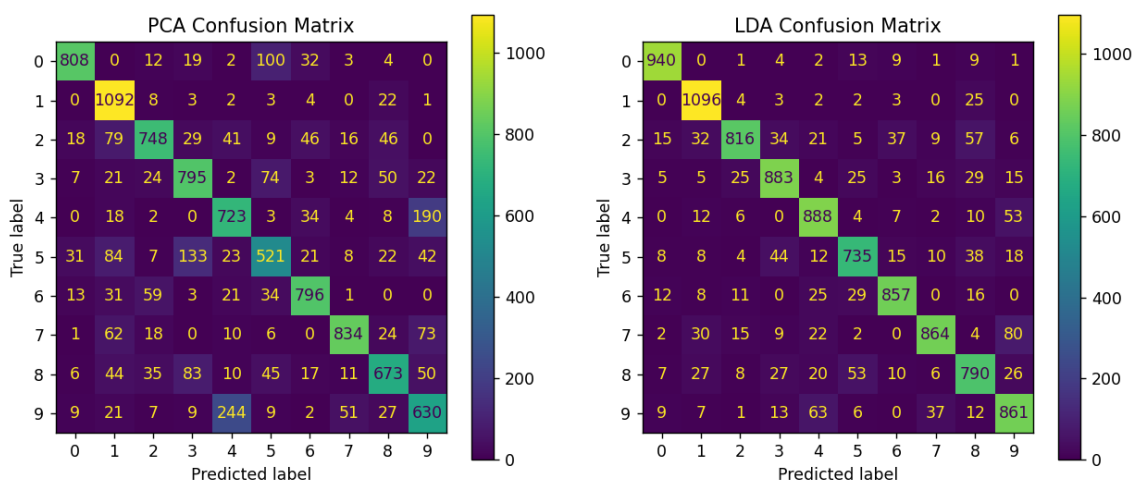
LDA Accuracy: 0.5413



For 9 dimensions, we get the following accuracies and confusion matrices:

PCA Accuracy: 0.762

LDA Accuracy: 0.873



Conclusion:

From this exercise we learn that increasing dimensionality reflects as a better performance for both LDA and PCA. We also learn that LDA performs better than PCA regardless of the dimension, as LDA is more effective for classification tasks on labeled data such as the mnist dataset, while PCA only preserves variance.